Assessment Menu: Teacher Guide

The Systems Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
GRAPH	Solve and interpret systems by graphing	A3, A4, E3, S2
SUBSTITUTE	Solve systems by substitution	A1, E1, E5, S3
ELIMINATE	Solve systems by elimination	A2, A6, E2, E6, S4
MODEL	Model situations with systems	A5, E4, S1

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 SUBSTITUTE	Solve by substitution: $y = 2x - 1$ and $3x + y = 14$	**Answer:** (3, 5) Substitute $2x - 1$ for y: $3x + (2x - 1) = 14$ $5x - 1 = 14$, so $5x = 15$ and $x = 3$ $y = 2(3) - 1 = 5$	2 pts: Substitutes $2x - 1$ for y correctly 1 pts: Solves for $x = 3$ 1 pts: Finds $y = 5$ 1 pts: States the solution as an ordered pair (3, 5)
A2 ELIMINATE	Solve by elimination: $x + y = 11$ and $x - y = 3$	**Answer:** (7, 4) Add the equations: $2x = 14$, so $x = 7$ Substitute: $7 + y = 11$, so $y = 4$ Check: $7 - 4 = 3$ ✓	2 pts: Adds the equations so y is eliminated 1 pts: $x = 7$ 1 pts: $y = 4$ 1 pts: Ordered pair (7, 4)
A3 GRAPH	Graph $y = x + 1$ and $y = -x + 5$ on the grid. Write the solution of the system.	**Answer:** (2, 3) $y = x + 1$: start at (0, 1), go up 1 and right 1 each time. $y = -x + 5$: start at (0, 5), go down 1 and right 1 each time. The lines cross at **(2, 3)**.	2 pts: Graphs $y = x + 1$ correctly 2 pts: Graphs $y = -x + 5$ correctly 1 pts: Solution (2, 3)
A4 GRAPH	The system $y = 2x + 3$ and $y = 2x - 4$ has how many solutions? Use the slopes and y-intercepts to explain.	**Answer:** No solution. The lines have the same slope (2) but different y-intercepts, so they are parallel and never cross. Both lines have slope 2. Their y-intercepts are different (3 and −4). Same slope and different intercepts means parallel lines, so there is **no solution**.	2 pts: Says no solution 1 pts: Notes the slopes are equal 1 pts: Notes the y-intercepts differ 1 pts: Concludes the lines are parallel and never intersect
A5 MODEL	Two numbers have a sum of 20 and a difference of 6. Write a system of equations and find the two numbers.	**Answer:** $x + y = 20$ and $x - y = 6$; the numbers are 13 and 7. Let x be the larger number and y the smaller. $x + y = 20$ and $x - y = 6$ Add: $2x = 26$, so $x = 13$; then $y = 7$. Check: $13 + 7 = 20$ and $13 - 7 = 6$ ✓	2 pts: Writes both equations correctly 2 pts: Solves the system correctly 1 pts: States both numbers: 13 and 7
A6 ELIMINATE	Solve by elimination: $2x + y = 7$ and $3x - y = 8$	**Answer:** (3, 1) Add the equations: $5x = 15$, so $x = 3$ Substitute: $2(3) + y = 7$, so $y = 1$ Check: $3(3) - 1 = 8$ ✓	2 pts: Adds to eliminate y 1 pts: $x = 3$ 1 pts: $y = 1$ 1 pts: Ordered pair (3, 1)

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 SUBSTITUTE	Solve by substitution: $x + 2y = 7$ and $3x - 4y = -9$. Check your answer in **both** equations.	**Answer:** (1, 3) $x = 7 - 2y$; then $3(7 - 2y) - 4y = -9$ $21 - 10y = -9$, so $y = 3$ and $x = 1$ Check: $1 + 6 = 7$ ✓ and $3 - 12 = -9$ ✓	2 pts: Isolates a variable in one equation 3 pts: Substitutes with parentheses and distributes correctly 2 pts: Solves for both variables: $y = 3$, $x = 1$ 3 pts: Checks (1, 3) in both equations
E2 ELIMINATE	Solve by elimination: $2x + 3y = -1$ and $3x + 5y = -2$	**Answer:** (1, −1) Multiply by 3 and −2: $6x + 9y = -3$ and $-6x - 10y = 4$ Add: $-y = 1$, so $y = -1$; then $2x - 3 = -1$, so $x = 1$	3 pts: Multiplies **both** equations (all terms) to create opposite coefficients 2 pts: Adds to eliminate one variable 3 pts: Finds $y = -1$ and $x = 1$ 2 pts: Ordered pair (1, −1) with a check or correct back-substitution
E3 GRAPH	Graph $2x + y = 4$ and $x - y = -1$ on the grid. Write the solution and check it in both equations.	**Answer:** (1, 2) Intercepts: (0, 4), (2, 0) and (0, 1), (−1, 0). The lines cross at **(1, 2)**; $2 + 2 = 4$ ✓ and $1 - 2 = -1$ ✓	3 pts: Graphs $2x + y = 4$ correctly 3 pts: Graphs $x - y = -1$ correctly 2 pts: Solution (1, 2) 2 pts: Checks in both equations
E4 MODEL	A school sold 120 tickets to a play. Student tickets cost $5 and adult tickets cost $8. The total was $735. Define variables, write a system, and find how many of each ticket were sold.	**Answer:** $s + a = 120$ and $5s + 8a = 735$; 75 student tickets and 45 adult tickets. s = student, a = adult: $s + a = 120$, $5s + 8a = 735$ $5(120 - a) + 8a = 735$ gives $600 + 3a = 735$, so $a = 45$ and $s = 75$	2 pts: Defines both variables 3 pts: Writes both equations correctly 3 pts: Solves correctly 2 pts: States the answer in words: 75 student, 45 adult
E5 SUBSTITUTE	Dev solved $y = 3x - 2$ and $4x + 2y = 16$ like this: $4x + 2(3x) - 2 = 16$ $10x - 2 = 16$ $x = \frac{9}{5}$ Find his mistake, then solve the system correctly.	**Answer:** He did not distribute the 2 to both terms of $(3x - 2)$. Correct: $x = 2$, $y = 4$. Should be $4x + 2(3x - 2) = 16$; the -2 must become -4. $10x - 4 = 16$, so $x = 2$ and $y = 4$: **(2, 4)**	3 pts: Identifies the mistake: 2 not distributed to −2 3 pts: Correct equation $10x - 4 = 16$ and $x = 2$ 2 pts: Finds $y = 4$ 2 pts: Ordered pair (2, 4)
E6 ELIMINATE	Use elimination to decide how many solutions each system has. **(a)** $4x - 6y = 10$ and $-2x + 3y = -5$ **(b)** $6x + 3y = 12$ and $2x + y = 9$	**Answer:** (a) infinitely many solutions (b) no solution (a) Second × 2: $-4x + 6y = -10$. Adding gives $0 = 0$ (always true): **infinitely many** (same line). (b) Second × −3: $-6x - 3y = -27$. Adding gives $0 = -15$ (never true): **none** (parallel lines).	2 pts: (a) Creates opposite coefficients correctly 2 pts: (a) Gets $0 = 0$ and concludes infinitely many 2 pts: (b) Creates opposite coefficients correctly 2 pts: (b) Gets $0 = -15$ and concludes no solution

			2 pts: Explains what each result means (same line, parallel lines)

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 MODEL	A boat travels 12 miles downstream in 2 hours. The return trip, 12 miles upstream, takes 3 hours. Let b be the boat’s speed in still water and c the current’s speed (miles per hour). **(a)** Write a system. **(b)** Find b and c. **(c)** How long would a 10-mile trip upstream take?	**Answer:** (a) $b + c = 6$ and $b - c = 4$ (b) $b = 5$ mph, $c = 1$ mph (c) 2.5 hours Downstream $12 \div 2 = 6$ mph; upstream $12 \div 3 = 4$ mph. (a) $b + c = 6$ and $b - c = 4$. (b) Add: $2b = 10$, so $b = 5$ and $c = 1$. (c) Upstream speed is 4 mph, so time $= \frac{10}{4} = 2.5$ hours.	4 pts: Finds the downstream and upstream speeds (6 and 4) 4 pts: Writes the system $b + c = 6$ and $b - c = 4$ 4 pts: Solves: $b = 5$, $c = 1$ with units 4 pts: Uses upstream speed of 4 mph for part (c) 4 pts: Correct time 2.5 hours with work shown
S2 GRAPH	Plant A is 2 cm tall and grows 2 cm per week. Plant B is 6 cm tall and grows 1 cm per week. Let x = weeks and y = height (cm). **(a)** Write an equation for each. **(b)** Graph both for weeks 0 to 6. **(c)** When are they the same height, and how tall? **(d)** After 8 weeks, which is taller and by how much?	**Answer:** (a) $y = 2x + 2$ and $y = x + 6$ (c) week 4, 10 cm (d) Plant A by 4 cm (a) $y = 2x + 2$ and $y = x + 6$. (b) Both lines pass through (4, 10); A starts at (0, 2), B at (0, 6). (c) Same height (**10 cm**) at **week 4**. (d) Week 8: A = 18 cm, B = 14 cm; A is taller by 4 cm.	4 pts: Both equations correct (2 pts each) 6 pts: Graphs both lines accurately (3 pts each) 4 pts: Intersection (4, 10) with meaning in context 3 pts: Computes both heights at week 8 (18 and 14) 3 pts: Concludes Plant A is taller by 4 cm
S3 SUBSTITUTE	Consider $3x + 2y = 8$ and $6x + 4y = c$. **(a)** Solve the first equation for y and substitute into the second. Simplify. **(b)** For what value of c are there infinitely many solutions? For which c are there none? **(c)** When c has that value, give two solutions.	**Answer:** (a) $16 = c$ (b) infinitely many when $c = 16$; no solution when $c \neq 16$ (c) for example (0, 4) and (2, 1) (a) $y = \frac{8 - 3x}{2}$; then $6x + 16 - 6x = c$, so **$16 = c$**. (b) $c = 16$ is always true: **infinitely many**. Any other c is false: **none**. (c) Same line, so any point of $3x + 2y = 8$: (0, 4), (2, 1).	3 pts: Solves the first equation for y correctly 4 pts: Substitutes and simplifies to $16 = c$ 4 pts: $c = 16$ gives infinitely many solutions (with reason) 3 pts: $c \neq 16$ gives no solution (with reason) 3 pts: First solution is on $3x + 2y = 8$ 3 pts: Second solution is on $3x + 2y = 8$
S4 ELIMINATE	The solution of $ax + 2y = 11$ and $3x + by = 7$ is (3, 1). **(a)** Find a and b. **(b)** Write the system and solve it by elimination to confirm (3, 1). **(c)** Maya says that changing b to 2 would give a system with no solution. Is she right? Explain using elimination.	**Answer:** (a) $a = 3$, $b = -2$ (b) $3x + 2y = 11$ and $3x - 2y = 7$ (c) Yes: subtracting gives $0 = 4$. (a) $3a + 2 = 11$ gives $a = 3$; $9 + b = 7$ gives $b = -2$. (b) Add $3x + 2y = 11$ and $3x - 2y = 7$: $6x = 18$, $x = 3$, $y = 1$ ✓ (c) With $b = 2$, subtracting gives $0 = 4$ (never true): Maya is right, **no solution**.	3 pts: $a = 3$ with substitution shown 3 pts: $b = -2$ with substitution shown 3 pts: Writes the correct system 3 pts: Eliminates correctly and finds (3, 1) 4 pts: Says Maya is right and shows $0 = 4$ (or equivalent) 4 pts: Explains that a false statement means no solution (parallel lines)